



Guidelines For Installation Of Innovative Standalone Solar Pumps

The Ministry of New and Renewable Energy (MNRE), India, has been supporting the installation of standalone solar pumps in the country under the Off-grid and Decentralised Solar PV Programme. In 2019-20, implementation of Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyaan (PM-KUSUM) started with a target to install 1.75 million standalone solar pumps by the year 2022.

The ministry has received representations from innovators claiming that by using different design/software for the solar pump, they can achieve performance better than specified by the ministry in a cost-effective manner. At present, under the MNRE scheme, solar pumps fulfilling the MNRE specifications can be installed; however, in order to promote innovation in technology, the ministry has decided to permit the installation of innovative standalone solar pumps in the test mode.

These guidelines are applicable for all Indian innovators/manufacturers/service providers who wish to install innovative standalone solar pumps in the country under schemes operated by the MNRE.

Call of Expression of Interest (EoI).

(a) The Ministry will call EoI from time to time for inviting applications from solar pump innovators claiming better performance on account of improved efficiency, cost-effectiveness, monitoring, other value addition, etc. Only those innovative products which are available for testing and field trials will be eligible for participation in the EoI.

(b) New technologies for which patent/IP related filings have been done (patent may not have been awarded) will also be eligible to participate in the EoI. Organisation that has filed the patent should be the sole applicant or lead partner in case of a consortium/JV.

(c) The elements of innovation in the context of improved performance of the solar pump need to be mentioned very clearly by the applicants in their response

to EoI. A cost-benefit analysis shall also be enclosed with the proposal.

Evaluation of applications. Applications will be evaluated by an evaluation committee constituted by the MNRE having technical and financial experts of the sector, including academic experts. Test report for the proposed solar pump will be submitted by the innovator along with EoI. However, if deemed necessary, the committee may recommend re-testing of performance of the pump at NISE or any other NABL accredited lab before allowing installation of such pumps in the field. In cases where the technology is in its nascent stage, a special method may be adopted to test the pumps in consultation with the innovator and NISE. The expenditure incurred on testing will be borne by the applicant.

Demonstration of innovation.

(a) On recommendations of the committee, the innovative technology will be allowed for demonstration in the field after getting consent from respective state implementing agencies (SIA) and beneficiary farmers by the innovator. The innovator will be allowed to install up to fifty solar pumps in different parts of the state/UT for demonstration purpose.

(b) For payment (subsidy and beneficiary share), annual maintenance contract (AMC), performance guarantee, etc, the innovators will be treated at par with other vendors selected through a competitive bidding process and awarded work for the installation of solar water pumps in the respective state/UT, and the innovators will not get any additional cost over and above the cost discovered through a tendering process.

Further, the innovators are required to deposit bank guarantee equivalent to payment made to them against the installation of solar pump, which will be encashed if the pump fails to meet at least the existing MNRE performance standards during the demonstration period of one year.

In addition, the innovators have to fill up indemnity bond against dam-

ages caused to farmer due to underperformance of the pump during the period of demonstration. On successful trial of the pump, the indemnity bond and bank guarantee would be released; however, the performance guarantee as applicable during the period of AMC will be retained by the SIA.

(c) The field demonstration will continue for at least one year from the date of installation of such pumps, and during that period the performance of the pump will be monitored on a real-time basis. Access to remote monitoring facility for the installed pumps shall be provided to NISE so that field data may be accumulated and evaluated.

(d) The innovator will submit every month a detailed performance report of the pump, including a comparative analysis with similar capacity pump specified by the MNRE, along with feedback from SIA and beneficiary farmers to the evaluation committee. MNRE/NISE will also conduct field inspection to check the performance as and when required.

(e) Installed innovative pumps shall meet at least the existing MNRE performance standards. However, only the innovations showing substantial improvement from the MNRE specifications may be considered for adoption. A comparative analysis of the performance of innovative technology vis-à-vis MNRE specifications will also be done by the evaluation committee.

(f) After detailed examination and deliberations, the evaluation committee may recommend for adoption of specifications of the innovative pump, with certain modifications, if required.

Adoption of technology by MNRE.

On recommendations of the evaluation committee, the MNRE may adopt the innovative technology and update the existing specification after having detailed stakeholders' consultation on the same. For innovations with the outstanding performance, innovators will get a certificate of appreciation from the MNRE. **EFY**

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